MINI PROJECT

Exploratory Data Analysis of Netflix Movies and TV Shows Dataset

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Abstract

This report presents an Exploratory Data Analysis (EDA) of the Netflix dataset, focusing on identifying trends, patterns, and insights related to content distribution. The dataset includes information on movies and TV shows such as genre, country, release year, and ratings. Data cleaning and preprocessing techniques were applied to handle missing values and improve data quality. Various visualizations were created to understand content trends, and key insights were derived. The analysis highlights Netflix's rapid growth, dominant content types, and global production patterns.

1. Introduction

Exploratory Data Analysis (EDA) is a fundamental step in data science that involves summarizing and visualizing data to extract meaningful insights. With the rise of streaming platforms, Netflix has become a leading provider of digital entertainment worldwide.

This project aims to analyze the Netflix dataset to understand content trends, growth patterns, and viewer-oriented features. By applying EDA techniques, we aim to uncover patterns that can help in understanding Netflix's content strategy.

2. Dataset Description

The dataset contains detailed information about Netflix movies and TV shows.

Key Features:

- Type (Movie / TV Show)
- Title
- Director
- Cast
- Country
- Date Added
- Release Year
- Rating
- Duration
- Genre
- Description

The dataset includes both categorical and numerical data, making it suitable for comprehensive analysis.

3. Before Data Cleaning

Initial analysis revealed the following issues:

- Missing values in:
  - date_added
  - year_added
  - month_added
- These columns are important for time-based analysis.
- Other columns were mostly complete and structured.

Conclusion:

Cleaning was necessary to ensure accurate analysis and visualization.

4. Data Cleaning and Preprocessing

The following steps were performed:

4.1 Handling Missing Values

- Filled missing values in date_added using the most frequent value.
- Recalculated year_added and month_added.

4.2 Data Type Conversion

- Converted date_added into datetime format.

4.3 Feature Engineering

- Created:
  - year_added
  - month_added

4.4 Data Consistency

- Ensured uniform formatting across dataset.

Result:

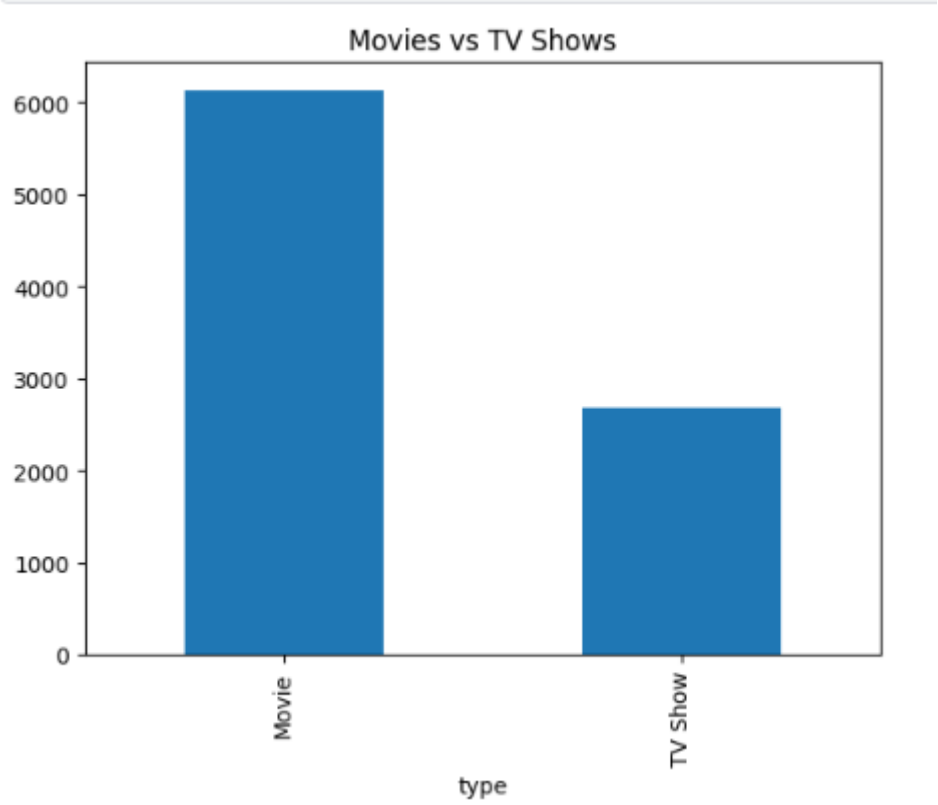
The dataset became clean, consistent, and ready for analysis.

5. Exploratory Data Analysis

EDA was performed using various visualization techniques to identify patterns and trends in the dataset.

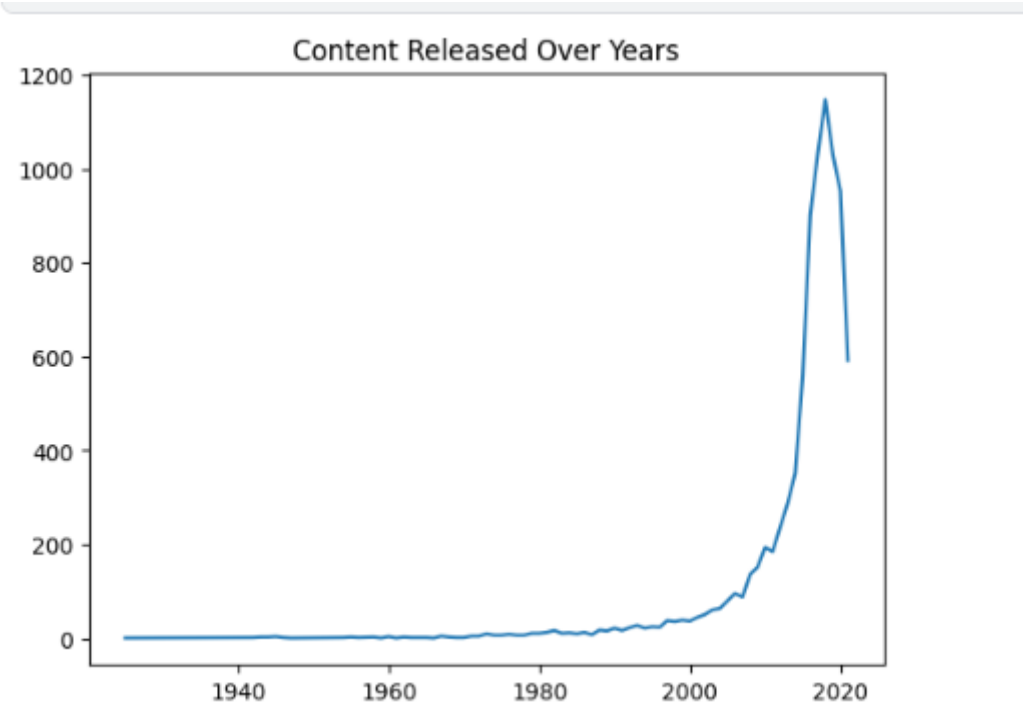
6. Visualizations and Analysis

Figure 1: Movies vs TV Shows



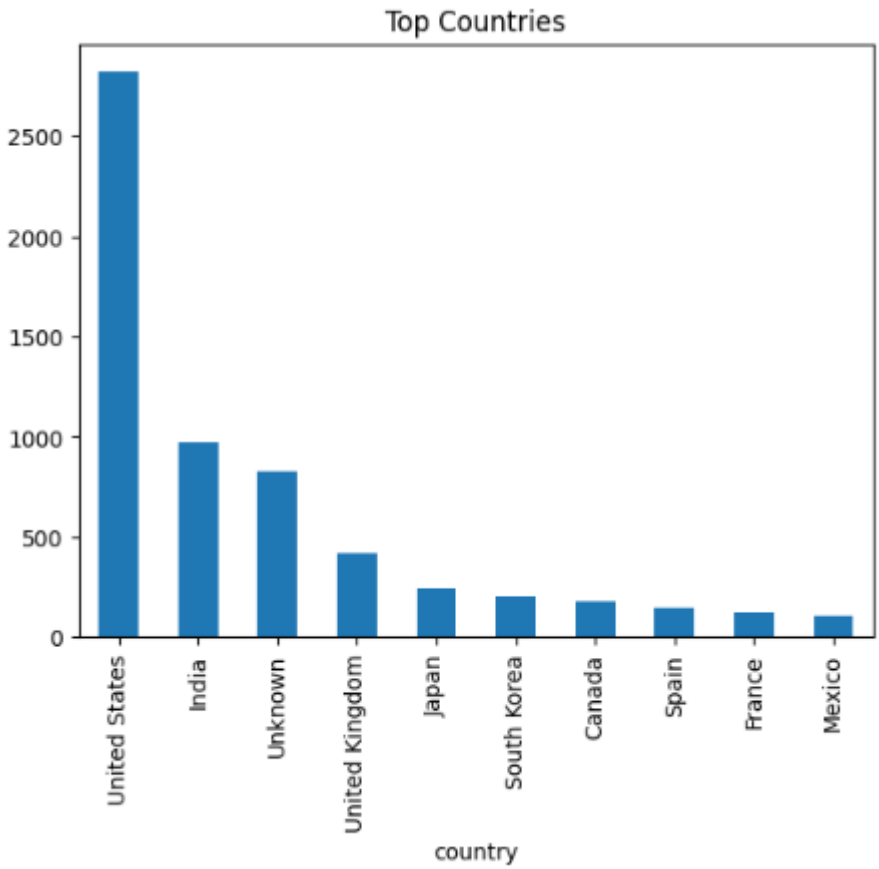
Insight:
Movies dominate the Netflix platform, indicating a higher focus on film content compared to TV shows.

Figure 2: Content Release Trend



Insight:
Content production increased significantly after 2010, showing rapid growth of Netflix.

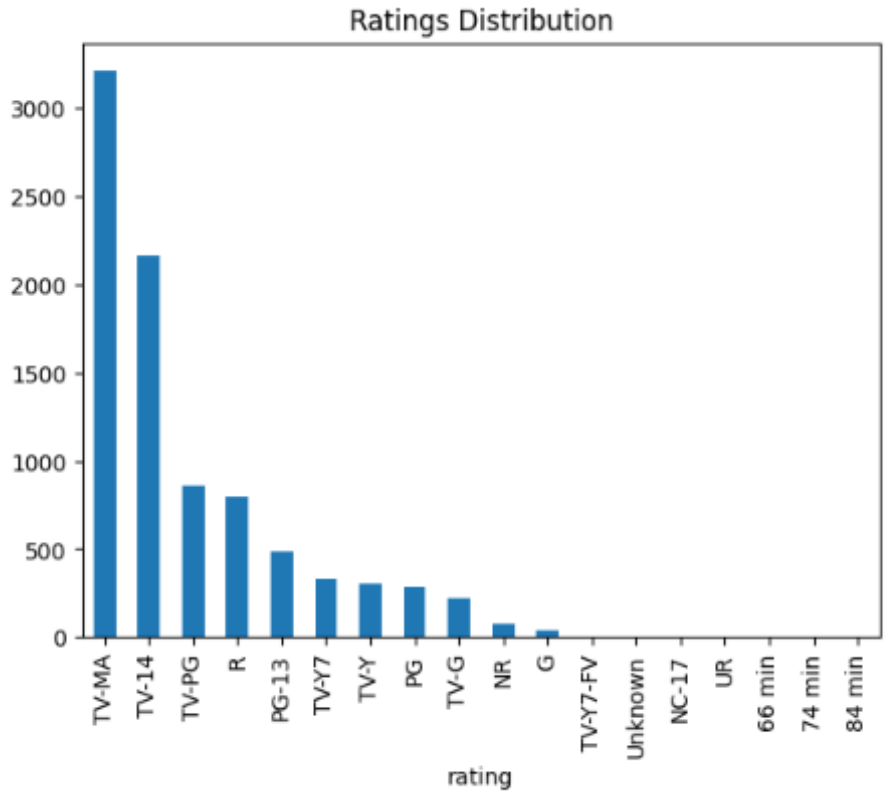
Figure 3: Top Content Producing Countries



Insight:

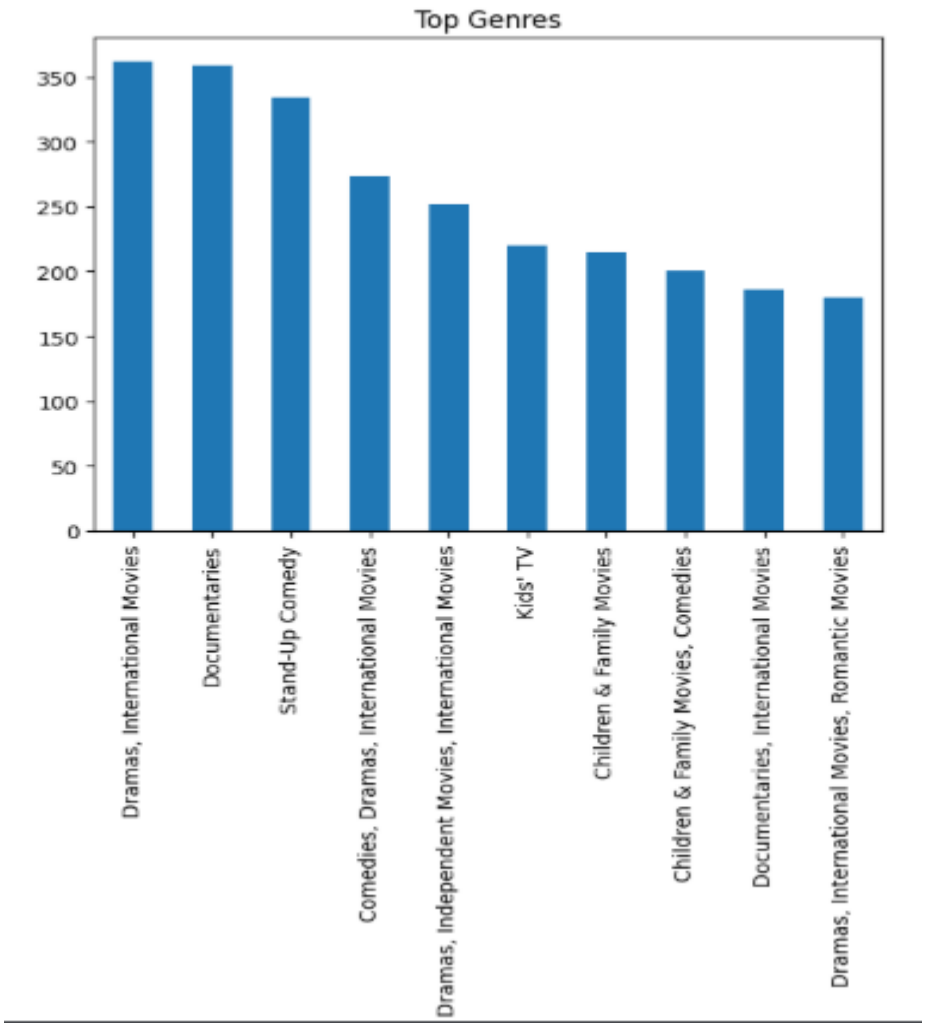
The United States produces the majority of content on Netflix.

Figure 4: Ratings Distribution



Insight:
Most content is rated TV-MA and TV-14, indicating a focus on mature audiences.

Figure 5: Genre Distribution



Insight:
Drama and Comedy are the most popular genres on Netflix.

7. Key Insights

- Netflix content has grown rapidly after 2010
- Movies dominate over TV Shows
- USA is the leading content producer
- Drama and Comedy are the most popular genres
- Content addition has increased significantly in recent years

8. Conclusion

The analysis highlights Netflix's rapid expansion and its evolution into a global content platform. The dataset reveals strong growth trends, diverse content offerings, and dominance of certain genres and countries.

EDA plays a vital role in understanding such datasets and provides valuable insights for decision-making.

9. Future Scope

Future improvements can include:

- Building recommendation systems
- Predicting content popularity using machine learning
- Performing sentiment analysis on user reviews
- Analyzing viewer engagement trends